Region based Segmentation

Region Growing

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region Growing

- $\max\{g(x,y)\} - \min\{g(x,y)\} <= 3$

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region splitting

	5	6	6	6	7	7	6	6	
R1	6	7	6	7	5	5	4	7	R2
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	4	7	
	0	0	0	0	2	2	5	6	
R3	1	1	0	1	0	3	4	4	R4
	1	0	1	0	2	3	5	4	

$$\mathbf{max}\{g(x,y)\} - \mathbf{min}\{g(x,y)\} <= 3$$

Region Splitting

- R1 and R3 satisfy homogeneity condition means all pixels in R1 satisfy threshold <=3 condition.

- R1 is 7 – 4 = 3
- R2 is 7 – 2 = 5
- R3 is 3 – 0 = 3
- R4 is 7 – 0 = 7

Region Splitting

	5	6	6	6	7	7	6	6
	6	7	6	7	5	5	4	7
R1	6	6	4	4	3	2	5	6
	5	4	5	4	2	3	4	6
	0	3	2	3	3	2	4	7
	0	0	0	0	2	2	5	6
R3	1	1	0	1	0	3	4	4
	1	0	1	0	2	3	5	4

R2

R4

Final Answer of Region splitting

Region Merging

- $\mathbf{max}\{g(x,y)\} - \mathbf{min}\{g(x,y)\}$ **<= 3**

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region Merging

- Check for four adjacent homogeneous regions
- Image after step 1

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region Split and Merge

Given Image Matrix

$$\begin{bmatrix} 10 & 12 & 200 & 205 \\ 11 & 13 & 198 & 202 \\ 9 & 10 & 50 & 55 \\ 8 & 9 & 52 & 54 \end{bmatrix}$$

Homogeneity condition

A region is homogeneous if:

$$\text{Max} - \text{Min} \leq 20$$

Step 1: Check the whole image

Whole matrix:

$$\begin{bmatrix} 10 & 12 & 200 & 205 \\ 11 & 13 & 198 & 202 \\ 9 & 10 & 50 & 55 \\ 8 & 9 & 52 & 54 \end{bmatrix}$$

- Minimum value = 8
- Maximum value = 205

$$205 - 8 = 197 > 20$$

So, the whole image is **not homogeneous**.

Hence, **split into 4 subregions**.

Step 2: Split into 4 regions

Region R_1

$$\begin{bmatrix} 10 & 12 \\ 11 & 13 \end{bmatrix}$$

- Min = 10, Max = 13

$$13 - 10 = 3 \leq 20$$

So, R_1 is homogeneous.

Region R_2

$$\begin{bmatrix} 200 & 205 \\ 198 & 202 \end{bmatrix}$$

- Min = 198, Max = 205

$$205 - 198 = 7 \leq 20$$

So, R_2 is homogeneous.

Region 3:

$$\begin{bmatrix} 9 & 10 \\ 8 & 9 \end{bmatrix}$$

✔ homogeneous

Region 4:

$$\begin{bmatrix} 50 & 55 \\ 52 & 54 \end{bmatrix}$$

✔ homogeneous

🔹 Step 4: Merging

Now check neighboring regions:

- Region 1 (≈10–13) and Region 3 (≈8–10)

  👉 Difference small → **merge**

✔️ New merged region:

$$\begin{bmatrix} 10 & 12 \\ 11 & 13 \\ 9 & 10 \\ 8 & 9 \end{bmatrix}$$

- Region 2 (≈200) → no merge
- Region 4 (≈50) → no merge

10	**12**	**200**	**205**
11	13	198	202
9	10	50	55
8	9	52	54